① Processos Estocásticos (Dia 04/11/2026)

Teo: Dado $\varphi \in \mathcal{L}^2_{loc}[0,T]$, seja $(\tau_n)_{n\in\mathbb{N}}$ localizante pra φ.

Defina $X_t^{(n)} = \int_0^t \varphi_s \mathbb{1}_{\{s\le\tau_n\}} dB_s$

$\exists (X_t)_{t\ge0}$ contínuo t.q. $\mathbb{P}\left(X_t = \lim_{n\to\infty} X_t^{(n)}\right) = 1 \ \forall t\in[0,T]$.

provai:

- Dado $n\ge m$, $\varphi_s \mathbb{1}_{\{s\le\tau_n\}} = \varphi_s \mathbb{1}_{\{s\le\tau_m\}}$ q.c. pra $s\le\tau_m$

$\Rightarrow$ $\int_0^t \varphi_s \mathbb{1}_{\{s\le\tau_n\}} dB_s = \int_0^t \varphi_s \mathbb{1}_{\{s\le\tau_m\}} dB_s$ q.c pra $t\le\tau_m$

$\Rightarrow$ $X_t^{(n)} = X_t^{(m)}$ q.c pra $t\le\tau_m$.

- Seja $N = \min\{n : \tau_n = T\}$.
- $(\tau_n)_{n\in\mathbb{N}}$ localizante $\Rightarrow \mathbb{P}(N<\infty) = 1$.
- Seja $\Omega_* = \{N<\infty\} \cap \{X^{(n)} \text{ é contínua } \forall n\in\mathbb{N}\}$
- $\mathbb{P}(\Omega_*) = 1$.
- Defina $X_t = X_t^{(N)}$, pra todo $\omega\in\Omega_*$.
- X é contínuo $\forall \omega\in\Omega_*$.
- $\forall \omega\in\Omega_*$, $X_t(\omega) = \lim_{n\to\infty} X_t^{(n)}$. ∎

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Def: Dado $\varphi \in \mathcal{L}^2_{loc}[0,T]$, seja X o processo dado pelo último teorema. Definimos

$$\int_0^t \varphi_s \, dB_s := X_t \quad \forall t \in [0,T].$$

Note que essa definição não depende da seq. localizante usada.

Martingais Locais

Def: $(M_t)_{t\geq 0}$ adaptado a $(\mathcal{F}_t)_{t\geq 0}$ é um martingal local se existe sequência não-decrescente $(\tau_n)_{n\in\mathbb{N}}$ de tempos de parada satisfazendo

(i) $\tau_n \longrightarrow +\infty$ q.c

(ii) $\forall n$, $M^{(n)}_t = M_{t\wedge\tau_n}$ é um martingal c.r.a $(\mathcal{F}_t)_{t\geq 0}$.

Teo: Dado $\varphi \in \mathcal{L}^2_{loc}[0,T]$, $\left(\int_0^T \varphi_s \, dB_s\right)_{t\in[0,T]}$ é um martingal local.

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Somas de Riemann

Teo: Dada $f:\mathbb{R}\to\mathbb{R}$ contínua e $t_i = iT/n$, $i=0,\dots,n$, então

$$\int_0^T f(B_t) = \mathbb{P}-\lim_{n\to+\infty} \sum_{i=0}^{n-1} f(B_{t_i})(B_{t_{i+1}} - B_{t_i})$$

Note que $(f(B_t))_{t\in[0,T]} \in \mathcal{L}^2_{loc}[0,T]$.

Integrais Gaussianas

Seja $\varphi:[0,T]\to\mathbb{R}$ contínua.

Seja $X_t = \int_0^t \varphi(s)\,dB_s$

$\Rightarrow (X_t)_{t\in[0,T]}$ é processo Gaussiano

$$\mathbb{E}(X_t)=0 \quad , \quad Cov(X_s X_t) = \int_0^{s\wedge t} \varphi(u)^2\,du$$

- Tome $t_{i,n} = iT/n$ e $t^*_{i,n} \in [t_{i,n}, t_{i+1,n}]$

$$\Rightarrow \int_0^T \varphi(s)\,dB_s = \mathbb{P}\text{-}\lim_{n\to\infty} \sum_{i=0}^{n-1} \varphi(t^*_{i,n})(B_{t_{i+1,n}} - B_{t_{i,n}})$$

(4) Fórmula de Itô

teorema fundamental do cálculo: $f \in C^1(\mathbb{R})$,

$$f(x) = f(0) + \int_0^x f'(y)\,dy \qquad df(x) = f'(x)dx + \frac{1}{2}f''(x)(dx)^2 + \frac{1}{3!}f'''(x)(dx)^3 + \dots$$

Teo (Fórmula de Itô) Dado $f \in C^2(\mathbb{R})$,

$$f(B_t) = f(B_0) + \int_0^t f'(B_s)\,dB_s + \frac{1}{2}\int_0^t f''(B_s)\,ds$$

prova: Suponha que f tem suporte compacto

$\Rightarrow \|f\|_\infty, \|f'\|_\infty, \|f''\|_\infty < \infty$.

- $f(y) - f(x) = (y-x)f'(x) + \frac{1}{2}(y-x)^2 f''(x) + r(x,y)$
- $r(x,y) = \int_x^y (y-s)(f''(s) - f''(x))\,ds$

  $\Rightarrow |r(x,y)| \le (y-x)^2 h(x,y)$.

  h unif. contínua, limitada, $h(x,x) = 0$.

- Dado $t > 0$, tome $t_i = it/n$.
- $f(B_t) - f(B_0) = \sum_{i=0}^{n-1} (f(B_{t_{i+1}}) - f(B_{t_i}))$

- $f(B_{t_{i+1}}) - f(B_{t_i}) = f'(B_{t_i})(B_{t_{i+1}} - B_{t_i}) + \frac{1}{2}f''(B_{t_i})(B_{t_{i+1}} - B_{t_i})^2 + r(B_{t_{i+1}}, B_{t_i})$

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$$A_n := \sum_{i=0}^{n-1} f'(B_{t_i})(B_{t_{i+1}} - B_{t_i}) \qquad (\text{quero } A_n \to \int_0^t f'(B_s)\,dB_s)$$

$$B_n := \frac{1}{2}\sum_{i=0}^{n-1} f''(B_{t_i})(B_{t_{i+1}} - B_{t_i})^2 \qquad (\text{quero } B_n \to \frac{1}{2}\int_0^t f''(B_s)\,ds)$$

$$C_n = \sum_{i=0}^{n-1} r(B_{t_{i+1}}, B_{t_i}) \qquad (\text{quero } C_n \to 0)$$

$$\Rightarrow f(B_t) - f(B_0) = A_n + B_n + C_n.$$

A_n

- f suporte compacto $\Rightarrow (f'(B_s))_{s\geq 0} \in \mathcal{H}^2$

(Soma de Riemann) $\Rightarrow \mathbb{P}\text{-}\lim_{n\to\infty} A_n = \int_0^t f'(B_s)\,dB_s.$

B_n

$$B_n = \frac{1}{2}\sum_{i=0}^{n-1} f''(B_{t_i})(t_{i+1}-t_i) + \underbrace{\frac{1}{2}\sum_{i=0}^{n-1} f''(B_{t_i})\left((B_{t_{i+1}}-B_{t_i})^2 - (t_{i+1}-t_i)\right)}_{\tilde{B}_n}$$

$s \mapsto f''(B_s)$ é contínua $\Rightarrow \frac{1}{2}\sum_{i=0}^{n-1} f''(B_{t_i})(t_{i+1}-t_i) \xrightarrow[q.c]{} \frac{1}{2}\int_0^t f''(B_s)\,ds.$

$$\mathbb{E}(\tilde{B}_n^2) = \frac{1}{4}\underbrace{\sum_{i\neq j} \mathbb{E}(\rightsquigarrow)}_{=0} + \frac{1}{4}\sum_{i=0}^{n} \mathbb{E}(\rightsquigarrow)$$

$$= \frac{1}{4}\sum_{i=0}^{n-1} \mathbb{E}\Big[\underbrace{f''(B_{t_i})^2}_{\leq \|f''\|_\infty^2}\underbrace{\left((B_{t_{i+1}}-B_{t_i})^2 - (t_{i+1}-t_i)\right)^2}_{\mathbb{E}(\cdot) = 2(t_{i+1}-t_i)^2}\Big]$$

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$$\leq \frac{1}{4}\sum_{i=0}^{n-1}\|f\|_\infty^2 2(t_{i+1}-t_i)^2 = \frac{2}{4}\|f\|_\infty^2 \sum_{i=0}^{n-1}\left(\frac{t}{n}\right)^2 = \frac{1}{2}\|f\|_\infty^2 \frac{t^2}{2n} \to 0$$

Logo $\tilde{B}_n \xrightarrow{L^2} \mathbb{E}(\tilde{B}_n) = 0$

$\mathbb{E}(\tilde{B}_n) = 0$. Logo $\tilde{B}_n \xrightarrow{L^2} 0$.

$\Rightarrow \tilde{B}_n \xrightarrow{\mathbb{P}} 0$.

C_n

$$\mathbb{E}[|C_n|] \leq \sum_{i=0}^{n-1} \mathbb{E}\left[(B_{t_{i+1}} - B_{t_i})^2 h(B_{t_{i+1}}, B_{t_i})\right]$$

$$\leq \sum_{i=0}^{n-1} \mathbb{E}\left[(B_{t_{i+1}} - B_{t_i})^4\right]^{1/2} \mathbb{E}\left[h(B_{t_{i+1}}, B_{t_i})^2\right]^{1/2}$$

$$= \sum_{i=0}^{n-1} \sqrt{3}(t_{i+1}-t_i)\, \mathbb{E}\left[h(B_{t_{i+1}}, B_{t_i})^2\right]^{1/2}$$

h unif. contínua $\Rightarrow$ Dado $\varepsilon > 0$, $\exists \delta > 0$ t.q $|x-y| < \delta \Rightarrow |h(x,y)| < \varepsilon$.

$$\Rightarrow \mathbb{E}\left[h(B_{t_{i+1}}, B_{t_i})^2\right]$$

$$= \mathbb{E}\left(h(B_{t_{i+1}}, B_{t_i})^2 \mid |B_{t_{i+1}} - B_{t_i}| < \delta\right) \mathbb{P}\left(|B_{t_{i+1}} - B_{t_i}| < \delta\right)$$
$$+ \mathbb{E}\left(h(B_{t_{i+1}}, B_{t_i})^2 \mid |B_{t_{i+1}} - B_{t_i}| \geq \delta\right) \mathbb{P}\left(|B_{t_{i+1}} - B_{t_i}| \geq \delta\right)$$

$$\leq \varepsilon^2 + \|h\|_\infty^2 \mathbb{P}\left(|B_{t_{i+1}} - B_{t_i}| \geq \delta\right)$$

$$\leq \varepsilon^2 + \|h\|_\infty^2 \mathbb{E}\left(|B_{t_{i+1}} - B_{t_i}|^2\right)/\delta^2 = \varepsilon^2 + \|h\|_\infty^2 \, t/n\delta^2$$

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$$\Rightarrow \mathbb{E}|C_n| \leq n\sqrt{3}\frac{t}{n}\left(\varepsilon^2 + \|h\|_\infty^2 \frac{t}{n\delta^2}\right)^{1/2}$$

$$\Rightarrow \limsup \mathbb{E}|C_n| \leq \sqrt{3}t\varepsilon \qquad \forall \varepsilon > 0$$

$$\Rightarrow C_n \xrightarrow{L^1} 0 \Rightarrow C_n \xrightarrow{\mathbb{P}} 0.$$

Logo

$$A_n \xrightarrow{\mathbb{P}} \int_0^t f'(B_s)dB_s$$

$$B_n \xrightarrow{\mathbb{P}} \frac{1}{2}\int_0^t f''(B_s)ds$$

$$C_n \xrightarrow{\mathbb{P}} 0$$

~~Tome subseq. t.q. os limites valem q.c.~~

Logo Formula de Itô vale pra t (e pra ~~uma~~ função compacta)

Mostre que vale simultâneo $\forall t \in \mathbb{Q}$

continuidade $\Rightarrow$ $\forall t \in \mathbb{R}$.

$\forall M \in \mathbb{N}$.

Pra f não compacta: Tome $f_M \in C^2(\mathbb{R})$ com suporte compacto t.q. $f_M(x) = f(x)$ $\forall |x| \leq M$.

$\tau_M = \min\{t \geq 0 : |B_t| \geq M\}$.

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$$f_M(B_t) = f_M(B_0) + \int_0^t f'_M(B_s)\,dB_s + \frac{1}{2}\int_0^t f''_M(B_s)\,ds \qquad \forall t$$

$f_M(B_s) = f(B_s)$ pra $s \leq \tilde{\tau}_M$

$$\Rightarrow \int_0^t f'_M(B_s)\,dB_s = \int_0^t f'(B_s)\,dB_s \quad \text{pra } t \leq \tilde{\tau}_M.$$

$\tau_M \longrightarrow \infty$ quando $M \to \infty$.

